

Ledger Epistemology and Regime-Closure Governance: An Operational Framework for Validation in Coupled Multi-Scale Systems

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Companion to the Regime-Closure Epistemology (RCE) theoretical paper, deposited in parallel.

Abstract. We present an operational framework for validation in complex empirical systems with coupled multi-scale dynamics. The framework comprises five integrated components: a regime architecture with explicit typology and coupling operators; frozen-parameter governance locking shared parameters across regime boundaries; sealed pre-registered predictions with cryptographic deposit hashes; a living falsification ledger preserving failed predictions with their failure modes documented; and external arbitration channels whose outcomes cannot be narratively reabsorbed by the validating apparatus. The framework was developed and field-tested over the period 2024–2026 within the Energy-Flow Cosmology (EFC) programme, which contains 163 deposited DOIs, 135 registered tests, 11 documented falsifications, and a public validation ledger in version 4.6. We argue that this governance machinery solves the central problem of validation in complex systems — the structural absorption of failed predictions through internal narrative coherence — which has limited the empirical traction of prior systemic philosophies (Bunge, Laszlo, Cilliers) and cybernetic frameworks (Ashby, Beer, Bateson). The underlying epistemic principle (Regime-Closure Principle, RCP) is convergent with the cybernetic tradition and is presented here as theoretical grounding rather than as primary novelty. The novelty claimed is the operational machine. The framework is logically and operationally independent of EFC’s empirical fate: whatever Stage-IV cosmological surveys return about EFC, the governance machinery is a separable contribution. We demonstrate the machine’s portability to medicine (M_0 – M_3 regime architecture), economics (E_0 – E_3), and metrology through pre-registered structural instantiations, and we declare five falsification conditions for the framework itself. Seven pre-registered cross-domain predictions with explicit thresholds and arbiters are deposited against this paper’s DOI.

Keywords: validation governance; falsification ledger; sealed prediction architecture; regime-closure; frozen-parameter governance; Energy-Flow Cosmology; cybernetics; cross-domain epistemology

1 Introduction: the validation governance problem

1.1 Internal narrative reabsorption as a structural failure mode

Empirical validation in coupled multi-scale systems has a persistent structural failure mode independent of any particular substantive theory. The failure mode is *internal narrative reabsorption*: when a field’s model produces a prediction that conflicts with data, the field’s internal interpretive resources are sufficiently flexible that the conflict can be reabsorbed without forcing model revision. The mechanism of reabsorption is well documented across domains: the discrepancy is relabelled as “residual”, “noise”, “model uncertainty”, “heterogeneity”, “exogenous shock”, “idiopathic case”, “systematic error of unknown origin”, or attributed to a mystery entity (“dark X”) postulated for the purpose of closing the ledger. The substantive theory is left undisturbed.

The pattern is structural, not malicious. Researchers do not choose to reabsorb failed predictions. The reabsorption is enabled by the absence of operational machinery that would prevent it. In particular, five operational components are typically absent:

- (i) explicit regime tagging of every prediction and every observable, so that cross-regime inference is forbidden by default;
- (ii) frozen-parameter governance, so that per-dataset re-tuning is recognised as model extension rather than legitimate fit;
- (iii) sealed pre-registered predictions, so that the result of the test is fixed before the data arrives;
- (iv) a living falsification ledger, so that failed predictions are preserved with their failure modes documented;
- (v) external arbitration, so that the outcome cannot be determined inside the apparatus that produced the prediction.

When all five components are present, narrative reabsorption is structurally blocked. When any are absent, reabsorption is the expected outcome regardless of the researcher's intentions. This paper claims, and demonstrates, that the joint imposition of these five components constitutes a workable operational framework for validation in coupled multi-scale systems, and that the framework has been field-tested over the period 2024–2026 in the Energy-Flow Cosmology (EFC) programme.

1.2 The cybernetic tradition's near-miss

The cybernetic tradition — Ashby's law of requisite variety (1956), Conant–Ashby's modelling theorem (1970), Beer's Viable System Model (1972), Bateson's logical types (1972), von Foerster's second-order cybernetics (1974), Friston's free-energy principle (2010) — identified the structural problem and developed the theoretical vocabulary for thinking about it. The cybernetic tradition's limitation was not theoretical but operational. It produced extensive philosophical literature about coupled multi-regime systems without producing falsifiable empirical programmes anchored in sealed predictions. The same is true of the explicitly systemic-philosophical programmes (Bunge's systemism, Laszlo's systems philosophy, Cilliers' work on complexity and postmodernism): rich theoretical content, weak operational anchoring.

The present paper inherits the theoretical grounding from cybernetics — the Regime-Closure Principle (RCP) developed in Section 3 is convergent with that grounding — and adds the operational machinery the cybernetic tradition lacked. The contribution claimed here is not in the principle but in its operationalisation.

1.3 The empirical case for the framework

The framework was developed within the Energy-Flow Cosmology (EFC) programme over 2024–2026. EFC is an alternative cosmological model proposing entropy-gradient gravitational response as the mechanism underlying observations conventionally attributed to dark matter and dark energy. The empirical fate of EFC as a cosmological model is independent of the present paper's thesis: the present paper is about the governance machinery EFC developed in the course of testing itself, not about whether EFC turns out to be the correct cosmology. This independence is load-bearing for the paper's argument and is made explicit in Section 4.3.

Concretely, the EFC corpus comprises 163 deposited DOIs on Figshare, 135 registered tests in a public validation ledger (current version 4.6), 11 explicitly documented falsifications preserved in the ledger with their failure modes, a Framework Atlas tracking 43 cosmological frameworks against 35 phenomena with 197 framework–phenomenon predictions, and an explicit four-layer

sync protocol (ORCID identity anchor, Figshare deposit layer, GitHub reproducer packages, public canonical narrative) that tracks drift between the layers as an operational health metric.

This is, to the author's knowledge, the most complete operational instantiation of validation governance in any contemporary alternative-physics programme. The claim is *etterprøvbart*: any prior programme that can demonstrate equivalent operational machinery (sealed predictions, deposit hashes, public falsification ledger with preserved failed predictions, frozen-parameter governance, external arbitration, machine-readable validation topology) is invited to document it. The point is not credit-establishment but mapping the boundary of operational completeness.

1.4 Cross-domain instantiations

The framework's portability is tested in three domains: medicine, economics, and metrology. In each case the paper presents a structural skeleton — regime typology, candidate coupling operators, candidate frozen parameters, candidate external arbiters — and a small set of pre-registered structural predictions that would be falsified by specific outcomes in specific data sources. The domain instantiations are not claimed to be empirically complete: they require domain experts to provide the substantive content. They are claimed to be operationally portable: the governance machinery applies without modification, only the substantive content differs.

The cross-domain claim is restricted to *homologous validation pathologies*, not to ontological identity across domains. The pattern of failure — single-regime inference applied to multi-regime systems producing mystery-entity postulation — is structurally the same; the mechanisms differ.

1.5 Independence from EFC's cosmological fate

A critical clarification before proceeding: the validation governance framework presented here is logically and operationally independent of EFC's empirical outcome as a cosmological theory. The decisive cosmological tests of EFC are pre-registered against Stage-IV surveys (Euclid DR1 in October 2026, Rubin LSST DP2 in mid-2026, DESI DR2/DR3 in 2026–2028, Simons Observatory × Euclid in 2027–2028). Whatever those surveys return, the governance machinery EFC has built in the course of preparing for them is a separable contribution.

If EFC is empirically validated by Stage-IV data, the governance machinery is vindicated by association — it was the operational discipline that made the prediction confrontation legible and honest. If EFC is empirically falsified, the governance machinery is arguably more vindicated — it produced the documented falsifications that closed the question rather than allowing them to be reabsorbed. The framework's claim does not require EFC to be correct; it requires EFC's validation apparatus to demonstrate that the joint imposition of (i)–(v) is operationally workable. That demonstration is already complete.

This independence is explicit so that readers do not need to evaluate EFC's cosmological claims to evaluate the present paper's thesis. The two are separable.

1.6 Structure of this paper

Section 2 presents the five-component governance architecture in operational detail. Section 3 grounds the architecture theoretically in the Regime-Closure Principle (RCP) and its cybernetic antecedents. Section 4 documents the EFC instantiation as the field-tested case, including the explicit independence claim. Section 5 develops cross-domain instantiations in medicine, economics, and metrology. Section 6 states the framework's own falsification conditions. Section 7 deposits seven pre-registered cross-domain predictions. Section 8 addresses anticipated objections. Section 9 concludes.

2 The five-component governance architecture

The framework comprises five integrated components. Each is necessary; together they are jointly sufficient to block internal narrative reabsorption. This section presents each component in operational detail, with explicit reference to its EFC implementation.

2.1 Regime architecture

A regime architecture is an explicit typology of the scales at which a complex system operates, together with the coupling operators that link adjacent regimes. The typology is constructed from three ingredients:

- **Characteristic scales.** A set of scales $\{\lambda_i\}$ separated by ratios $\lambda_{i+1}/\lambda_i \geq K$ for some scale-separation constant $K > 1$. The separation may be spatial, temporal, energetic, or organisational, but it must be *declared and justified*.
- **Characteristic dynamics.** Each scale λ_i admits a distinct effective theory $\mathcal{T}_i$ with its own degrees of freedom, conservation laws, and equations of motion. The effective theories are not reducible to one another except via explicit coarse-graining or refinement operators.
- **Coupling operators.** Adjacent regimes are linked by operators $\mathcal{C}_{i,i+1}$ that map the state space of $\mathcal{T}_i$ into the state space of $\mathcal{T}_{i+1}$. These operators are not identity maps; they are where buffering, emergence, and screening occur, and they must be specified explicitly — not left implicit in the model.

EFC implementation. The cosmological regime architecture is L_0 – L_3 : L_0 (early universe, linear), L_1 (transition), L_2 (structure formation, weak nonlinearity), L_3 (galactic and sub-galactic, fully nonlinear). The architecture is documented in the Foundations paper (DOI 10.6084/m9.figshare.31135597), formalised in RCMP (DOI 10.6084/m9.figshare.31222900), and refined into the L_0 – L_3 specification (DOI 10.6084/m9.figshare.31112536). Coupling operators are explicit: $T(a)$ for background-to-perturbation transition, $\Theta(\rho)$ for density saturation across the L_2 – L_3 boundary, $\Sigma_{\text{eff}}(z)$ for the perturbation-to-lensing coupling.

Why it is necessary. Without an explicit regime architecture, cross-regime inference happens by default. An observable in one regime is silently used to constrain a mechanism in another regime, with no explicit coupling operator to justify the inference. The result is systematic ambiguity: an apparent inconsistency may reflect failure of the model, failure of the coupling, or failure of the measurement, and the three cannot be disentangled. Explicit regime tagging makes the three disentangleable in principle, even when they cannot be disentangled in any given dataset.

2.2 Frozen-parameter governance

A parameter θ that appears in multiple regimes is governed by the constraint

$$\theta|_{R_i} = \theta|_{R_j} \quad \forall i, j,$$

unless an explicit declaration is made that θ is regime-local. Per-dataset re-tuning of a frozen parameter is treated as model extension and must be declared as such; it cannot be silently absorbed into the validation procedure as “fit”.

EFC implementation. The Global Parameter Registry (documented in the Validation Ledger v4.6, public at the canonical location) lists every parameter with its operational regime and its freeze status. Failed re-tuning attempts are preserved as explicit falsifications: the Variant H entropy-gradient MCMC against DESI DR2 BAO + $f\sigma_8$ + $H(z)$ + SNIa returned $\Delta\text{AIC} = +4.05$ favouring ΛCDM under joint frozen parameters; this is preserved in the ledger as a documented null, not relabelled as a “feature” or absorbed via per-dataset adjustment.

Why it is necessary. Without frozen-parameter governance, the most common failure mode in multi-regime modelling is silent reabsorption of failed predictions via per-regime tuning. The model appears to fit every dataset by re-tuning its parameters per dataset, but the cross-dataset consistency that would constitute genuine empirical support is structurally invisible. Frozen-parameter governance is the empirical-side analogue of the Conant–Ashby modelling closure theorem (Section 3.2): the validation procedure must contain a model of the system, and that model cannot be silently extended.

2.3 Sealed prediction architecture

A sealed prediction is a numerical prediction deposited on a public archive with a cryptographic deposit hash before the corresponding data is available. The deposit timestamp and hash fix the prediction beyond the possibility of subsequent revision. Pass/fail criteria, arbiter datasets, and falsifier conditions are part of the seal and cannot be adjusted after the fact.

EFC implementation. Ten sealed predictions are currently deposited across the L0–L3 architecture:

- P1: lensing crossover at $z \approx 0.44$, sign change in $\Sigma_{\text{eff}}(z)$ across $z \in [0.3, 0.6]$ tomographic bins. DOI 10.6084/m9.figshare.32037990. Arbiter: Euclid DR1 (October 2026).
- P2: $f\sigma_8(z = 0.7) = 0.430$ vs $\Lambda\text{CDM } 0.449$. DOI 10.6084/m9.figshare.32013156. Arbiter: DESI DR2 full-shape RSD.
- P3: $E_G^{\text{EFC}}/E_G^{\text{GR}} = 1.086 \pm 0.012$ at $k_c \approx 0.05 h/\text{Mpc}$, $z_{\text{eff}} \approx 0.5$. DOI 10.6084/m9.figshare.32023788. Arbiter: $\text{SO} \times \text{Euclid DR2}$.
- P4: Rubin DP2 cosmic shear $S_8 = 0.847 \pm 0.015$. DOI 10.6084/m9.figshare.32013738. Arbiter: Rubin LSST DP2.
- P5–P8: structural and field-equation-level seals (Horndeski no-go, background sign lemma, three-window H_0 decomposition, bar-instability Π_{EFC} criterion).

Each seal includes a SHA-256 hash on the prediction artefact, a deposit timestamp on Figshare, and explicit pass-window and kill-condition thresholds.

Why it is necessary. Without sealed predictions, the result of a test is not fixed at the time of testing. A field can adjust its predictions, its thresholds, or its arbiter datasets in response to results, with the adjustment hidden in the natural drift of research. Sealed predictions block this by making the entire test specification cryptographically immutable. Beer’s Viable System Model identified this requirement in 1972 (the “algedonic channel” that bypasses internal narrative); the present paper adds the cryptographic implementation.

2.4 Living falsification ledger

A falsification ledger is a public, machine-readable record of every test the framework has produced, with explicit status tags (passed, failed, planned, deprecated), failure modes documented for each failed test, and version history preserving the trajectory. Falsified predictions are not deleted; they are preserved with their failure mode and the date of falsification.

EFC implementation. The Validation Ledger is currently at version 4.6, public at the canonical location supertedai.github.io/EFC/public/. It records 135 registered tests across five categories (48 physics_test, 21 consistency_check, 16 phenomenological, 28 framework_constraint, 22 planned_pipeline). Of the resolved tests, 104 survive and 11 are explicitly falsified. The 11 falsifications are preserved with failure modes:

- EFC v1 scalar-index model: monotonic radial boost incompatible with observed rotation-curve diversity; abandoned in favour of regime-dependent EBE framework.
- EFC-BIC v0.1 bar-instability prediction: AUC 0.217 (anti-correlated with bar truth) on SPARC; track closed, Π_{EFC} pivoted to candidate disk-state proxy.
- Original ISW cancellation metric $\mathcal{C}$: spurious tomographic gradient from mixed-channel implementation; superseded by self-consistent ODE growth-channel uniform $A_{\text{ISW}} \approx 0.89$.
- Naive GRAV-to-cosmology mapping: produced $\mu > 1$ (wrong sign); closed by EFC Relativistic Action derivation.
- Poisson-sector $\mu > 1$ growth boost: disfavoured by RSD; architecture revised to require $\mu < 1$.

- Four further falsifications documented with full failure modes.

The ledger is machine-readable (JSON-LD), version-controlled, and synchronised across four layers (ORCID identity anchor, Figshare deposit, GitHub reproducer packages, public canonical narrative) with explicit drift tracking between layers.

Why it is necessary. Without a falsification ledger, survivorship bias is structural. A field’s published record reflects its surviving predictions, not its full prediction history. Failed predictions disappear from the apparent track record, and the field appears to have a better predictive record than it does. Beer’s 1972 VSM and Bateson’s logical-types theory both pointed to this as the central pathology of self-validating systems; the ledger is the operational fix.

2.5 External arbitration channels

An external arbiter is a data source whose result is determined outside the apparatus that produced the prediction, and whose confrontation window is pre-registered before the data is available. The arbiter must be (i) external to the validating institution, (ii) operating on a protocol fixed before the prediction was made, and (iii) capable of returning a result that resolves the prediction in either direction.

EFC implementation. Stage-IV cosmological surveys serve as arbiters: Euclid DR1 (operated by ESA, protocol fixed 2019–2024, data release October 2026) for P1 lensing crossover; Rubin LSST DP2 (operated by Vera Rubin Observatory, protocol fixed 2018–2024) for P4 cosmic shear; DESI DR2/DR3 (operated by DESI collaboration, protocol fixed 2020–2024) for P2 RSD; Simons Observatory $\times$ Euclid (joint protocol 2024–2026) for P3 E_G slip.

None of these arbiters is operated by the EFC apparatus. Each has its own pre-registered analysis protocols, calibration pipelines, and data-release schedules. Each can return a result that resolves EFC’s predictions in either direction with no possibility of narrative reabsorption by the EFC apparatus.

Why it is necessary. Internal-only validation is structurally insufficient: sufficiently complex systems can narratively absorb arbitrarily discrepant results when the arbitration apparatus is part of the system. External arbiters bypass this absorption capacity by construction. This is Beer’s algedonic channel made cryptographically literal: the result arrives from outside the system’s interpretive boundary, and the system cannot rewrite its own arrival.

2.6 Why all five are necessary

The five components are not redundant. Each addresses a distinct mode of narrative reabsorption, and removing any one re-opens that mode:

- Without (i) regime architecture: cross-regime inference is silent and ambiguous; failed predictions can be reabsorbed by relabelling the regime.
- Without (ii) frozen parameters: per-dataset re-tuning is silent; failed predictions can be reabsorbed by silent parameter adjustment.
- Without (iii) sealed predictions: the prediction itself drifts; failed predictions can be reabsorbed by retrospective revision.
- Without (iv) falsification ledger: survivorship bias operates; failed predictions are reabsorbed by deletion from the record.
- Without (v) external arbitration: the validating apparatus determines its own validation; failed predictions are reabsorbed by reinterpretation.

Joint imposition is the operational core of the framework. The theoretical justification — why these five are exactly the necessary conditions — is the content of Section 3.

3 Theoretical grounding: the Regime-Closure Principle

The five-component governance architecture is empirically motivated — it addresses the documented failure modes of single-regime inference in coupled multi-scale systems. This section provides the theoretical grounding: an operational principle, the Regime-Closure Principle (RCP), that characterises the conditions for valid validation in such systems. The principle is presented here as grounding for the framework; the framework is the primary contribution and stands or falls with its operational performance.

3.1 Statement

Regime-Closure Principle (RCP). Let S be a complex system composed of $N \geq 2$ coupled subsystems $\{S_i\}_{i=1}^N$, where each S_i operates at characteristic timescale τ_i or characteristic scale λ_i , satisfying a scale-separation condition $\tau_{i+1}/\tau_i \geq K$ for some constant $K > 1$. Let $\{S_i\}$ be coupled by inter-regime operators $\{\mathcal{C}_{i,j}\}$. Let V be a validation procedure applied to a model M of S . Then V is a *valid* procedure for M if and only if the following five conditions are jointly satisfied:

- (a) **Regime locality.** Each prediction π_i of M pertaining to subsystem S_i is tested against observables $\mathcal{O}_i$ generated within S_i . Cross-regime inference requires explicit coupling operators $\mathcal{C}_{i,j}$ that are themselves derived or independently constrained.
- (b) **Variety closure.** The validation pipeline V has variety $V(V) \geq V(S)/V(B)$ where $V(S)$ is the state-space variety of S and $V(B)$ is the buffering variety. (Direct consequence of Ashby’s law applied to validation rather than control.)
- (c) **Modelling closure.** V contains a structural model of every S_i and of every active coupling operator $\mathcal{C}_{i,j}$. (Direct consequence of the Conant–Ashby theorem applied to validation: a regulator must model its regulated system; a validator must model its validated system.)
- (d) **Frozen-parameter governance.** Every parameter θ that appears in two or more regimes is locked, $\theta|_{R_i} = \theta|_{R_j} \ \forall i, j$, unless declared regime-local.
- (e) **External arbitration.** V employs at least one channel of sealed, pre-registered prediction whose outcome is determined externally to the validation pipeline.

When (a)–(e) jointly fail, the symptom is the postulation of mystery entities (the “dark X” phenomenon) whose role is to close the ledger that the validation procedure has been forbidden from closing mechanically.

3.2 Status: principle, not theorem

We label the result a *principle*, not a *theorem*. A theorem in the strict mathematical sense is derived from axioms by formal proof. RCP is not of that kind: it is a characterisation of necessary and sufficient operational conditions for validation procedures in coupled multi-regime systems, where “valid” is defined relative to an empirical standard (production of replicable, falsifiable predictions whose failure modes are documented) rather than from first principles. RCP takes Conant–Ashby’s modelling theorem and Ashby’s variety inequality as load-bearing components — those are formally proven — but RCP itself is an operational principle, in the same epistemic category as the Regime-Consistent Measurement Principle (RCMP, DOI 10.6084/m9.figshare.31222900) and Entropy-Bounded Empiricism (EBE, DOI 10.6084/m9.figshare.31222903).

3.3 Cybernetic substrate

The principle is convergent with prior work in the cybernetic tradition, which supplies the formally-proven theorems on which the operational conditions are grounded. The five cybernetic results below correspond one-to-one with conditions (a)–(e) of RCP, with the exception of frozen-parameter governance (d) which is empirically motivated rather than theoretically derived.

- **Ashby’s Law of Requisite Variety (1956).** A regulator R of a system S subject to disturbance D can stabilise S only if the variety of R satisfies $V(R) \geq V(D)/V(B)$ where $V(B)$ is buffering variety. Applied to validation rather than control, this is condition (b) of RCP.
- **Conant–Ashby Theorem (1970).** Every good regulator of a system must contain a model of that system. Applied to validation: a validation pipeline must contain a structural model of each regime of the system being validated. This is condition (c) of RCP.
- **Bateson’s logical types (1972).** Confusion of logical types is the structural origin of a broad class of pathologies. Applied here: confusion of regime level is the structural origin of mystery-entity postulation.
- **Beer’s Viable System Model (1972).** Algedonic channels are necessary for survival of complex systems because internal coherence absorbs internal signals. Applied to validation: sealed pre-registered predictions function as algedonic channels for scientific theories. This is condition (e) of RCP.
- **Friston’s free-energy principle (2010).** Any self-organising system minimises variational free energy. The EFC Cross-Domain Bridge Equations paper (DOI 10.6084/m9.figshare.31940547) establishes that EFC’s gradient-flow dynamics is structurally identical to FEP.

The cybernetic tradition supplied the theorems; this paper’s contribution is the operationalisation that the cybernetic tradition did not produce.

3.4 Mystery-entity diagnosis

The mystery-entity symptom — “dark X” across domains — is the diagnostic signature that one or more RCP conditions are being violated. The diagnosis is structural, not pejorative. A field producing a mystery entity is not committing an error; it is operating under a measurement architecture that cannot close its ledger by mechanical means, and the mystery entity is the bookkeeping device that closes it.

We distinguish three categories of mystery entity, which the framework treats as structurally homologous but ontologically distinct:

- **Empirically characterised, ontologically unidentified.** Dark matter as currently constituted: precisely measured gravitational signature, unknown particle content. The empirical characterisation is real; the ontology is not.
- **Descriptive residual categories.** Medical “idiopathic” or “essential” labels: categories whose content is exactly their failure of mechanism-attribution. The empirical characterisation is weaker; the category is closer to a placeholder than to a measured signature.
- **Explicit modelling residuals.** Knightian uncertainty in economics; “internal variability” in climate ensembles: residuals honestly declared as outside the model’s scope.

The framework claims these three categories share a structural homology: each arises when single-regime inference is applied to a multi-regime system. They are not claimed to be

ontologically identical. A precisely characterised gravitational signature is not the same epistemic object as a clinical placeholder label. What is shared is the structural cause of the gap they fill.

This distinction matters for the dissolution prediction: RCP-compliant validation in cosmology may reveal that dark matter is partially a real entity (a particle yet to be detected) *and* partially a regime-closure artefact. RCP-compliant validation in medicine is more likely to reveal that idiopathic categories are predominantly regime-closure artefacts, because the descriptive-residual category has weaker empirical anchoring to begin with. The framework does not predict that all mystery entities are artefacts. It predicts that the fraction explained by regime closure is presently underestimated across these domains.

4 Energy-Flow Cosmology as the field-tested case

4.1 Brief description

Energy-Flow Cosmology (EFC) is an alternative cosmological framework in which gravity exhibits regime-dependent response governed by entropy-gradient dynamics. The mathematical core is an effective gravitational coupling $\mu(a)$, an effective lensing amplitude $\Sigma(k, z)$, and a gravitational slip $\eta(k, z)$, all derived from a relativistic action with explicit regime-transition operators. A narrow “survival valley” in the (μ, Σ, η) parameter space encodes the joint constraints from CMB lensing, growth-rate data, and weak-lensing surveys: $\mu \in [0.93, 0.96]$, $\Sigma \in [1.03, 1.07]$, $\eta \approx 1.10$.

The Foundations paper (DOI 10.6084/m9.figshare.31135597) provides the architectural overview. The full corpus comprises 163 deposited DOIs documenting field equations, perturbation theory, validation tests, falsifications, atlas comparisons, and cross-domain bridges.

For the present paper, EFC’s substantive cosmological content is secondary. What matters is that EFC was tested under all five RCP conditions simultaneously over a 14-month period, and that the test apparatus accumulated the operational machinery the present paper claims is the load-bearing contribution.

4.2 Operational track record

The track record demonstrating the framework’s operability:

- **135 registered tests** across five categories. The breakdown (48 physics tests, 21 consistency checks, 16 phenomenological, 28 framework constraints, 22 planned pipelines) illustrates the variety-matching breadth condition (b) of RCP requires.
- **104 surviving tests** after data confrontation, 11 documented falsifications.
- **11 explicit falsifications preserved** with failure modes and version-control trace. No falsification has been silently removed; each is permanent in the ledger.
- **Four-layer sync protocol** tracking drift between ORCID identity (S1, root anchor), Figshare deposit (S2, golden source), GitHub reproducer packages (S3), and public canonical narrative (S4, currently Validation Ledger v4.6). Layer inclusion is $S1 \subset S2 \subset S3 \subset S4$; deviations are tracked as drift events.
- **Framework Atlas** with 43 cosmological frameworks $\times$ 35 phenomena $\times$ 197 framework–phenomenon predictions, providing the variety-matching infrastructure for multi-framework discrimination.
- **Ten sealed numerical predictions** (P1–P8 plus two further) with SHA-256 deposit hashes, pre-registered pass-windows and falsifier thresholds, and external arbiters assigned to each.
- **Confrontation calendar** with arbiter releases pre-dated: Rubin DP2 (Jul–Sep 2026), Euclid DR1 (October 21, 2026), DESI DR2 (2026), DESI DR3 (2027–2028), Simons Observatory $\times$ Euclid (2027–2028).

The functional content of these enumerations is that the framework has demonstrated, over 14 months, that the joint imposition of (i) regime tagging, (ii) frozen parameters, (iii) sealed predictions, (iv) falsification ledger, and (v) external arbitration is workable. Whether the EFC predictions themselves survive Stage-IV is a separate question.

4.3 Independence claim

The framework's claim is logically separable from EFC's empirical fate. Three explicit scenarios:

1. **If EFC is validated by Stage-IV data:** the governance machinery is vindicated by association — it was the operational discipline that allowed the prediction confrontation to be legible. The framework gains an empirical success case.
2. **If EFC is partially falsified** (some sealed predictions pass, others fail): the machinery is more vindicated, not less, because it produced the documented failures rather than reabsorbing them. The framework gains a documented case of machine-led falsification.
3. **If EFC is fully falsified** (all major sealed predictions fail): the machinery is most vindicated, because it produced the cleanest possible empirical closure — a cosmological alternative was sealed, tested, and falsified with no possibility of internal narrative reabsorption. The framework gains its strongest possible operational demonstration.

In none of these scenarios does the present paper's thesis depend on EFC being correct. The thesis is that the joint imposition of (i)–(v) constitutes a workable operational discipline; the EFC corpus is the demonstration that the discipline is workable. The demonstration is complete regardless of EFC's empirical outcome.

5 Cross-domain instantiations

This section instantiates the framework in three domains other than cosmology: medicine, economics, and metrology. AI alignment is discussed briefly. In each case the paper presents a structural skeleton, not an empirically complete instantiation. Domain experts are required to provide the substantive content.

5.1 Medicine: the M0–M3 architecture

Scope and stringency caveat. The medical instantiation in this subsection is a structural skeleton, not an empirical validation of EFC-equivalent stringency. The author holds no medical credentials. The regime architecture proposed below is intended as a draft to be corrected by domain experts. The pre-registered predictions in Section 7.1 are structural inferences from the framework rather than clinical hypotheses derived from independent biomedical evidence.

Mystery entities. Medicine generates an unusually clean set of descriptive-residual categories: *idiopathic* (cause unknown), *essential* (cause unknown, restated), *functional* (cause unknown, located in function rather than structure), *psychosomatic* (cause unknown, located in psyche), *autoimmune of unknown trigger*, *central sensitisation*, *neuroinflammation of unknown origin*. The prevalence and burden of conditions in these categories (long COVID, ME/CFS, fibromyalgia, IBS, atopic multi-morbidity, treatment-resistant depression, idiopathic autoimmunity, chronic inflammatory states) is substantial and growing.

Diagnosis under the framework. Medicine systematically violates RCP conditions (a)–(c) by organisational design. Specialist silos — cardiology, endocrinology, immunology, gastroenterology, psychiatry, neurology — map onto physiological subsystems that are coupled by feedback loops on overlapping timescales. A patient with dysregulation distributed across HPA-axis, autonomic balance, gut microbiota, immune tone, and circadian rhythm presents to one silo at a time; each silo measures within its own regime and either finds nothing actionable or treats locally. The systemic coupling is unmeasured.

The M0–M3 architecture. Mirroring the cosmological L_0 – L_3 structure, we propose four medical regimes separated by characteristic timescales spanning from molecular kinetics to organism-level behaviour.

Regime	Timescale	Active observables
M_0 molecular / genomic	seconds–minutes	enzyme kinetics, signalling cascades, gene expression
M_1 cellular / tissue	minutes–hours	inflammatory markers, hormone pulses, autocrine signalling
M_2 organ / system	hours–days	HRV, cortisol rhythm, insulin sensitivity, autonomic balance
M_3 organism / behaviour	days–years	sleep, social contact, nutrition, identity, habit

Inter-regime coupling operators include the HPA axis ($M_2 \leftrightarrow M_3$), gut–brain axis ($M_1 \leftrightarrow M_3$), allostatic load (McEwen 1998, formally $M_2 \leftrightarrow M_3$ buffer exhaustion), and chronobiological coupling ($M_0 \leftrightarrow M_3$).

Frozen-parameter governance candidates. A patient’s allostatic state, autonomic tone, and circadian phase are parameters that should be locked across M_0 – M_3 during diagnosis. At present they are typically measured by no specialist and tracked by none. An RCP-compliant clinical pipeline would require these to be characterised once at baseline and held fixed across all subsequent specialist measurements within the same diagnostic episode.

External arbiters. Pre-registered RCTs and longitudinal cohort studies with locked analysis protocols (e.g. ClinicalTrials.gov registration before recruitment) serve as algedonic channels. The existence of partial governance infrastructure in medicine (ClinicalTrials.gov, CONSORT, PRISMA, OSF pre-registration) is a strength relative to economics; the gap is the absence of regime-locking and frozen-parameter discipline.

Predicted dissolution. Under the framework, conditions currently classified as idiopathic are predicted to be predominantly cross-regime dysregulations visible only when measurement architecture respects regime structure. Pre-registered predictions are in Section 7.1.

5.2 Economics: the E0–E3 architecture

Mystery entities. Economics generates mystery entities of its own: *exogenous shocks* (cause located outside the model boundary by fiat), *irrational actors* (residual variance attributed to agents), *animal spirits*, *liquidity preference*, unexplained *risk premia*, *Knightian uncertainty*.

Diagnosis. DSGE modelling and its descendants typically assume separability between micro-foundations, credit dynamics, and macro-aggregate stabilisation. This is an explicit single-regime architecture imposed on a system with strong nested feedback on disparate timescales. Goodhart’s Law (“when a measure becomes a target, it ceases to be a good measure”) is an RCP violation: a measurement optimised within one regime drives the system into another regime where the measurement is no longer valid.

The E0–E3 architecture. The economic regime structure parallels the medical and cosmological cases:

Regime	Timescale	Active observables
E_0 transactional	seconds–days	order flow, payment settlement, daily liquidity
E_1 business cycle	months–years	inflation, employment, credit growth
E_2 institutional	years–decades	regulatory regimes, financial architecture
E_3 structural / civilisational	decades–centuries	demographic transition, technological paradigm

Coupling operators are well-documented but rarely explicitly modelled: credit buffering ($E_0 \leftrightarrow E_1$), institutional path-dependence ($E_1 \leftrightarrow E_2$), and Minsky-style instability accumulation ($E_2 \leftrightarrow E_3$).

Predicted dissolution. A substantial fraction of variance currently attributed to “exogenous shocks” is predicted to be internally generated by E_0 – E_1 or E_1 – E_2 coupling that exceeds the buffering capacity of the lower regime. Operationally, this is the Minsky financial-instability hypothesis stated as an RCP corollary. Pre-registered predictions in Section 7.2.

5.3 Metrology: the instrument-coupling case

Metrology is structurally distinct because the validation procedure V is itself coupled to the system S being analysed: the instrument and the measurand are not independent. The Regime-Bound Measurement paper (DOI 10.6084/m9.figshare.31564123) and the Regime-Locked Measurement paper (DOI 10.6084/m9.figshare.31833076) develop this case in detail.

The mystery entity is *systematic error of unknown origin*. Under the framework’s diagnosis, such errors are RCP violations: the instrument is operating outside its calibrated regime, or the measurand has crossed a regime boundary the instrument was not designed to track, or instrument and measurand are coupled through a buffer that absorbs the signal of interest.

The metrological instantiation produces concrete operational guidance: every measurement campaign must (i) tag the regime of the instrument, (ii) tag the regime of the measurand, (iii) declare any coupling between them, and (iv) freeze instrument-side parameters across the campaign.

5.4 AI alignment: the deployment-distribution case

In machine learning, mystery entities include *emergent capability*, *deceptive alignment*, *specification gaming*, and *distribution shift*. The training regime and the deployment regime are distinct L_i in the RCP sense. Validation that measures the former and predicts the latter without explicit coupling operators is an RCP violation, formally identical to the KT3b cosmology case (DOI 31963821). The Validity-Aware AI paper develops this instantiation in detail (DOI 31122970).

6 Falsification conditions for the framework

The framework is itself falsifiable. Five conditions, jointly or severally, falsify it:

- (F1) **Single-regime success in a multi-regime domain.** If a domain governed by genuinely coupled multi-regime dynamics achieves validated predictive success using strictly single-regime inference — no explicit coupling operators, no frozen-parameter governance, no external sealed predictions — and the success is replicated independently. Operational threshold: predictive $R^2 > 0.5$ on out-of-sample data spanning two or more regimes, achieved without any regime coupling structure in the model.
- (F2) **Regime-closed validation systematically failing.** If domains adopting framework-compliant validation produce *worse* predictive performance than single-regime alternatives across multiple instantiations. Operational threshold: in three or more independent domain implementations, framework-compliant validation yields larger out-of-sample prediction error than single-regime baseline at $p < 0.05$.
- (F3) **Mystery entities surviving regime closure.** If a domain adopts framework-compliant validation in full and the mystery entity that motivated the implementation survives intact rather than dissolving into explicit operators or becoming independently measurable. Operational threshold: surviving entities must be independently measurable within five years of framework adoption; if not, the dissolution claim is falsified for that domain.
- (F4) **Cross-domain isomorphism failure.** If predictions structurally derived in one domain do not survive translation to another domain following the cross-domain mapping.

- (F5) **Variety inequality non-binding.** If validation pipelines with $V(V) \ll V(S)/V(B)$ systematically produce equally good predictive performance, Ashby’s variety constraint applied to validation is falsified, and the framework’s cybernetic grounding is undermined.

7 Pre-registered cross-domain predictions

The following predictions are deposited as sealed pre-registrations against this paper’s DOI. Each prediction has an explicit observable, quantitative threshold, external arbiter, and falsification criterion. Deposit serves as the algedonic channel: internal narrative reabsorption is structurally blocked.

7.1 Medicine

P-MED-1 (M_2 – M_3 coupling visibility). In a cohort of patients with currently-idiopathic chronic conditions (long COVID, ME/CFS, fibromyalgia; $n \geq 500$ per condition), simultaneous frozen measurement of M_2 parameters (allostatic load index, 24-hour HRV autonomic balance, cortisol rhythm slope) across all participants will reveal cluster structure with effect size $d \geq 0.5$ that does not appear in conventional single-specialist diagnostic workflows. Falsifier: clusters do not emerge, or emerge with $d < 0.3$. Arbiter: independent replication in two or more cohorts. Window: ≤ 5 years.

P-MED-2 (Cross-regime intervention superiority). Therapeutic protocols addressing cross-regime coupling explicitly (combined M_2/M_3 intervention with frozen-parameter tracking) will outperform best-current-practice single-specialist intervention on idiopathic chronic conditions, relative effect size $\geq 1.4\times$. Falsifier: relative effect size $< 1.1\times$. Arbiter: pre-registered RCT, $n \geq 200$. Window: ≤ 7 years.

P-MED-3 (Variety-closure diagnostic precision). A diagnostic protocol enforcing simultaneous measurement across M_0, M_1, M_2, M_3 regimes with locked baselines will reduce the “idiopathic” classification rate by $\geq 40\%$ relative to current standard-of-care. Falsifier: reduction $< 15\%$. Arbiter: side-by-side specialty-clinic comparison. Window: ≤ 5 years.

7.2 Economics

P-ECON-1 (Coupling-aware crisis prediction). A macroeconomic forecasting protocol enforcing E_0 – E_1 coupling explicitly (real-time credit buffering metrics held as frozen parameters) will outperform state-of-the-art DSGE benchmarks on out-of-sample crisis-onset forecasting, relative false-negative reduction $\geq 30\%$ at matched false-positive rate. Falsifier: reduction $< 10\%$. Arbiter: out-of-sample evaluation on post-deposit crisis episodes. Window: ≤ 10 years.

P-ECON-2 (Goodhart-rate regime dependence). Goodhart-rate (decoupling of a metric from its underlying construct under policy targeting) will scale with regime distance: Goodhart-rate $\propto |E_i - E_j|$ where targeting acts at E_i and construct is defined at E_j . Falsifier: no significant correlation with regime distance. Arbiter: historical analysis of pre-registered policy episodes. Window: ≤ 3 years.

7.3 Metrology

P-MET-1 (Cross-regime calibration deficit). Residual systematic error in precision measurement campaigns where instrument-regime and measurand-regime are distinct will scale as $\sigma_{\text{sys}} \propto V(S)/V(V)$. Falsifier: σ_{sys} independent of variety mismatch within precision. Arbiter: meta-analysis of precision-measurement campaigns with documented cross-regime structure. Window: ≤ 5 years.

7.4 Cross-domain meta-prediction

P-META-1 (Isomorphism transfer). Predictions derived structurally in one domain will admit translation into a structurally equivalent prediction in another domain following the cross-domain mapping, and the translated prediction will be empirically supported (or falsified) in correlation with the source prediction. If domain-1 prediction passes and domain-2 prediction

fails (or vice versa) while domain conditions are otherwise similar, the isomorphism claim is weakened. Window: ≤ 10 years.

8 Anticipated objections

8.1 “This is just rebranded multi-scale modelling.”

Multi-scale modelling exists under headings of renormalisation, hierarchy theory, and multi-physics simulation. The framework is more specific: it addresses *validation* rather than modelling, it imposes frozen-parameter governance across scales, and it requires external sealed arbitration. Multi-scale modelling typically does none of these.

8.2 “This is just cybernetics restated.”

The framework is convergent with the cybernetic tradition — Ashby, Conant–Ashby, Beer, Bateson, von Foerster, Friston — as Section 3.3 documents. What is new is the operational machinery: sealed predictions with deposit hashes, a living falsification ledger preserving failed predictions with failure modes, frozen-parameter governance enforced across regime boundaries, and explicit external arbitration channels. None of the cybernetic precursors operationalised these jointly. The cybernetic tradition produced theorems and philosophical positions; the framework provides an operational machine.

8.3 “This is Bunge/Laszlo/Cilliers redux.”

Twentieth-century systemic-philosophy programmes (Bunge’s systemism, Laszlo’s general systems philosophy, Cilliers’ work on complexity and postmodernism) attempted comparable syntheses and produced substantial philosophical literature without producing falsifiable empirical programmes. The objection is whether the present framework converges to the same outcome. The answer is structural: those programmes lacked the four operational components the framework inherits from EFC — sealed numerical predictions with deposit hashes, a living falsification ledger, frozen-parameter governance, and external arbitration channels. Bunge, Laszlo, and Cilliers had philosophical positions; the framework has an operational machine. The distinction is empirical: a philosophy of science cannot be falsified by data; an operational governance framework instantiated in a specific predictive programme can be. If the present framework’s sealed predictions never resolve either way, the distinction is not operationally demonstrated and the framework defaults to the same epistemic status as its predecessors.

8.4 “You are gluing fields you don’t have credentials in.”

The author has no credentials in medicine, economics, or metrology. The pre-registered predictions in Section 7 are structural inferences from the framework, not domain-expert claims. The appropriate response from domain experts is to evaluate whether the proposed regime architecture matches the field’s actual coupling structure, and if not, to provide the corrected architecture. The framework stands or falls with the corrected architectures, not with the author’s draft versions.

8.5 “The dark-X framing collapses distinct epistemic categories.”

The framing distinguishes three categories (Section 3.4): empirically characterised but ontologically unidentified (dark matter); descriptive residual labels (idiopathic); explicit modelling residuals (Knightian uncertainty). The claim is structural homology, not ontological identity. A precisely characterised gravitational signature is not the same epistemic object as a clinical placeholder label; what is shared is the structural cause of the gap each fills in the field’s ledger.

8.6 “Universal pretensions in any field require extraordinary evidence.”

Acknowledged. The framework makes no universal claim. It predicts homologous validation pathologies in domains with coupled multi-regime structure. Domains without genuine scale separation, or with measurement architectures that already enforce regime-locked inference, are

explicitly outside scope. The cross-domain predictions in Section 7 are restricted to domains where the multi-regime structure is independently established.

9 Conclusion

The contribution of this paper is the joint imposition of five operational components for validation in coupled multi-scale systems: explicit regime architecture with declared coupling operators, frozen-parameter governance across regime boundaries, sealed pre-registered predictions with cryptographic deposit hashes, a living falsification ledger preserving failed predictions with failure modes documented, and external arbitration channels whose outcomes cannot be narratively reabsorbed.

The five components are convergent with theoretical content from the cybernetic tradition — Ashby’s variety law, Conant–Ashby’s modelling theorem, Beer’s algedonic channels, Bateson’s logical types, Friston’s free-energy principle. The framework’s novelty is not in the theoretical grounding but in the operationalisation, which the cybernetic tradition did not produce. The Regime-Closure Principle (Section 3) is offered as the operational characterisation of necessary and sufficient conditions; it is convergent with prior work and is labelled a principle, not a theorem.

The framework was developed within the Energy-Flow Cosmology programme over 2024–2026. The EFC corpus comprises 163 deposited DOIs, 135 registered tests, 11 documented falsifications, a public validation ledger in version 4.6, a 43-framework atlas, and ten sealed predictions against Stage-IV cosmological surveys arriving in 2026–2028. The framework’s empirical adequacy is independent of EFC’s cosmological fate: whether EFC is empirically validated or falsified by Stage-IV, the operational machinery EFC built is a separable contribution, and the demonstration that the machinery is workable is already complete.

The framework’s portability is tested in pre-registered structural instantiations in medicine (M_0 – M_3), economics (E_0 – E_3), and metrology, with seven sealed cross-domain predictions deposited in Section 7. Five falsification conditions for the framework itself are stated in Section 6. The author makes no claim that the framework is correct, only that it is sufficiently operationalised to be tested, and declares the tests by which it will be.

If the framework holds, then validation in coupled multi-scale systems admits an operational discipline that has been absent across multiple empirical domains, and a substantial fraction of contemporary “mystery entity” postulation is a remediable structural pathology rather than a generational mystery. If the framework fails, the failure will be documented in the ledger and the corpus will be revised accordingly. Either outcome advances the operational discipline of validation in coupled multi-regime systems.

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Version 1.0. This document is a deposit-ready preprint. The predictions in Section 7 are pre-registered against the DOI assigned at deposit. Falsification entries will be added to the EFC Validation Ledger and reflected in subsequent versions. The companion theoretical paper (Version A: Regime-Closure Epistemology) is deposited in parallel and cross-references this document as its operational implementation. The two papers together cover the theoretical claim and the operational machinery; either may be cited independently.